# GENERAL SCIENCE: CHEMISTRY FUNDAMENTALS

**ATOM AND PERIODIC IDENTITY → BOND AND FORCE LADDER → STOICHIOMETRIC ACCOUNTING RAIL → STATE AND SOLUTION MATRIX → ACID BASE BUFFER CONTROL → ELECTROCHEMICAL TWO-CELL MAP**

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00

## ATOM AND PERIODIC IDENTITY MAP

ATOM AND PERIODIC IDENTITY MAP · PROTONS + NEUTRONS · ATOMIC NUMBER · ELEMENT IDENTITY · MASS NUMBER · DIFFERENT COMPARISON AXES · PERIODIC TABLE · ATOMIC NUMBER + RECURRING PROPERTIES

NUCLEUS → protons + neutrons

ATOMIC NUMBER → protons → element identity → MASS NUMBER → protons + neutrons → ISOTOPE / ISOBAR / ISOTONE → different comparison axes

**CONCEPT / ARCHITECTURE**
- NUCLEUS → protons + neutrons
- ATOMIC NUMBER → protons → element identity

**MECHANISM / APPLICATION**
- MASS NUMBER → protons + neutrons
- ISOTOPE / ISOBAR / ISOTONE → different comparison axes

**READINESS / STATUS LIMIT**
- PERIODIC TABLE → atomic number + recurring properties
- NUCLEUS → protons + neutrons

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PERIODIC TABLE → atomic number + recurring properties.

01

STAGE 01

## BOND AND FORCE LADDER

BOND AND FORCE LADDER · ELECTRON TRANSFER + ION ATTRACTION · SHARED ELECTRON PAIR · DELOCALISED ELECTRONS · HYDROGEN BOND · VAN DER WAALS · WEAKER INTERPARTICLE FORCES · DO NOT COLLAPSE BOND AND FORCE CATEGORIES

IONIC → electron transfer + ion attraction

COVALENT → shared electron pair → METALLIC → delocalised electrons → HYDROGEN BOND / VAN DER WAALS → weaker interparticle forces

**STARTING CONDITION / INPUT**
- IONIC → electron transfer + ion attraction
- COVALENT → shared electron pair

**SCIENTIFIC OR ENGINEERING MECHANISM**
- METALLIC → delocalised electrons
- HYDROGEN BOND / VAN DER WAALS → weaker interparticle forces

**OUTPUT / FAILURE BOUNDARY**
- TRAP → do not collapse bond and force categories
- IONIC → electron transfer + ion attraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → do not collapse bond and force categories.

02

STAGE 02

## STOICHIOMETRIC ACCOUNTING RAIL

STOICHIOMETRIC ACCOUNTING RAIL · WRITE SPECIES · BALANCE EQUATION · CONVERT SUPPLIED QUANTITIES · COMPARE STOICHIOMETRIC REQUIREMENT · IDENTIFY LIMITING REAGENT · THEORETICAL PRODUCT · YIELD, ONLY WITH SUPPLIED DATA

WRITE SPECIES → BALANCE EQUATION → CONVERT SUPPLIED QUANTITIES → MOLES → COMPARE STOICHIOMETRIC REQUIREMENT → IDENTIFY LIMITING REAGENT → THEORETICAL PRODUCT → ACTUAL / THEORETICAL → YIELD, only with supplied data

**STARTING CONDITION / INPUT**
- WRITE SPECIES → BALANCE EQUATION
- CONVERT SUPPLIED QUANTITIES → MOLES

**SCIENTIFIC OR ENGINEERING MECHANISM**
- COMPARE STOICHIOMETRIC REQUIREMENT
- IDENTIFY LIMITING REAGENT → THEORETICAL PRODUCT

**OUTPUT / FAILURE BOUNDARY**
- ACTUAL / THEORETICAL → YIELD, only with supplied data
- WRITE SPECIES → BALANCE EQUATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACTUAL / THEORETICAL → YIELD, only with supplied data.

03

STAGE 03

## STATE AND SOLUTION MATRIX

STATE AND SOLUTION MATRIX · ARRANGEMENT + MOBILITY + SEPARATION · PHASE CHANGE · PHYSICAL IDENTITY RETAINED · MOLECULAR/IONIC DISPERSION · AMOUNT ON A STATED BASIS · EQUILIBRIUM LIMIT UNDER STATED CONDITIONS

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| SOLID / LIQUID / GAS → arrangement + mobility + separation | SOLUTION → molecular/ionic dispersion | SOLUBILITY → equilibrium limit under stated conditions |
| PHASE CHANGE → physical identity retained | CONCENTRATION → amount on a stated basis | SOLID / LIQUID / GAS → arrangement + mobility + separation |

**SYSTEM / DEVICE / INSTITUTION**
- SOLID / LIQUID / GAS → arrangement + mobility + separation
- PHASE CHANGE → physical identity retained

**DECISIVE TECHNICAL DISTINCTION**
- SOLUTION → molecular/ionic dispersion
- CONCENTRATION → amount on a stated basis

**STATUS / UNIT / EVIDENCE LIMIT**
- SOLUBILITY → equilibrium limit under stated conditions
- SOLID / LIQUID / GAS → arrangement + mobility + separation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOLID / LIQUID / GAS → arrangement + mobility + separation.

04

STAGE 04

## ACID BASE BUFFER CONTROL LOOP

ACID BASE BUFFER CONTROL LOOP · ACID DONATES H+ <-> BASE ACCEPTS H PH · LOGARITHMIC ACTIVITY MEASURE · ACIDIC/BASIC CHARACTER REDUCED · CONSUMES LIMITED ADDED ACID/BASE · SALT SOLUTION · NEUTRAL DEPENDS ON IONS

**CONCEPT / ARCHITECTURE**
- ACID DONATES H+ <-> BASE ACCEPTS H pH → logarithmic activity measure
- NEUTRALISATION → acidic/basic character reduced

**MECHANISM / APPLICATION**
- BUFFER → consumes limited added acid/base
- SALT SOLUTION → acidic / basic / neutral depends on ions

**READINESS / STATUS LIMIT**
- ACID DONATES H+ <-> BASE ACCEPTS H pH → logarithmic activity measure
- NEUTRALISATION → acidic/basic character reduced

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SALT SOLUTION → acidic / basic / neutral depends on ions.

05

STAGE 05

## ELECTROCHEMICAL TWO-CELL MAP

ELECTROCHEMICAL TWO-CELL MAP · GALVANIC CELL · CHEMICAL ENERGY TO ELECTRICAL ENERGY · ELECTROLYTIC CELL · ELECTRICAL ENERGY DRIVES CHEMISTRY · APPLIED ROUTES · SIGN TRAP · CELL TYPE CHANGES SIGNS, NOT REACTION LOCATIONS

OXIDATION → anode

REDUCTION → cathode → GALVANIC CELL → chemical energy to electrical energy → ELECTROLYTIC CELL → electrical energy drives chemistry

**STARTING CONDITION / INPUT**
- OXIDATION → anode
- REDUCTION → cathode

**SCIENTIFIC OR ENGINEERING MECHANISM**
- GALVANIC CELL → chemical energy to electrical energy
- ELECTROLYTIC CELL → electrical energy drives chemistry

**OUTPUT / FAILURE BOUNDARY**
- CORROSION / PLATING / EXTRACTION → applied routes
- SIGN TRAP → cell type changes signs, not reaction locations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIGN TRAP → cell type changes signs, not reaction locations.

06

STAGE 06

## KINETICS VERSUS EQUILIBRIUM

KINETICS VERSUS EQUILIBRIUM · RATE AXIS · COLLISIONS + ACTIVATION ENERGY · ALTERNATIVE LOWER-ENERGY PATHWAY · EQUILIBRIUM AXIS · FORWARD RATE EQUALS REVERSE RATE · FASTER APPROACH IN BOTH DIRECTIONS · NO SHIFT

RATE AXIS → collisions + activation energy

CATALYST → alternative lower-energy pathway → EQUILIBRIUM AXIS → forward rate equals reverse rate → CATALYST → faster approach in both directions

**CONCEPT / ARCHITECTURE**
- RATE AXIS → collisions + activation energy
- CATALYST → alternative lower-energy pathway

**MECHANISM / APPLICATION**
- EQUILIBRIUM AXIS → forward rate equals reverse rate
- CATALYST → faster approach in both directions

**READINESS / STATUS LIMIT**
- NO SHIFT → equilibrium position unchanged
- RATE AXIS → collisions + activation energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EQUILIBRIUM AXIS → forward rate equals reverse rate.

07

STAGE 07

## ORGANIC AND POLYMER FAMILY TREE

ORGANIC AND POLYMER FAMILY TREE · FUNCTIONAL GROUPS · ADDITION POLYMER · NO SMALL-MOLECULE ELIMINATION · CONDENSATION POLYMER · SMALL MOLECULE ELIMINATED · THERMOPLASTIC <-> THERMOSET · REHEATING BOUNDARY

FUNCTIONAL GROUPS → alcohol

ADDITION POLYMER → no small-molecule elimination → CONDENSATION POLYMER → small molecule eliminated → THERMOPLASTIC <-> THERMOSET → reheating boundary

**CONCEPT / ARCHITECTURE**
- FUNCTIONAL GROUPS → alcohol
- ADDITION POLYMER → no small-molecule elimination

**MECHANISM / APPLICATION**
- CONDENSATION POLYMER → small molecule eliminated
- THERMOPLASTIC <-> THERMOSET → reheating boundary

**READINESS / STATUS LIMIT**
- HYDROGEL → cross-linked hydrophilic network
- FUNCTIONAL GROUPS → alcohol

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FUNCTIONAL GROUPS → alcohol | aldehyde | ketone | acid | amine | ester.

08

STAGE 08

## MATERIALS STRUCTURE-PROPERTY WHEEL

MATERIALS STRUCTURE-PROPERTY WHEEL · TETRAHEDRAL NETWORK · LAYERED STRUCTURE · DISTINCT CARBON ARCHITECTURES · COMPOSITION + PROCESSING ALTER PROPERTIES · SAME ELEMENT CAN YIELD DIFFERENT MATERIAL BEHAVIOUR

**CONCEPT / ARCHITECTURE**
- DIAMOND → tetrahedral network
- GRAPHITE → layered structure

**MECHANISM / APPLICATION**
- GRAPHENE / FULLERENE / NANOTUBE → distinct carbon architectures
- ALLOY → composition + processing alter properties

**READINESS / STATUS LIMIT**
- RULE → same element can yield different material behaviour
- DIAMOND → tetrahedral network

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → same element can yield different material behaviour.

09

STAGE 09

## FUEL CONVERSION FAULT TREE

FUEL CONVERSION FAULT TREE · OXIDATION + USEFUL HEAT + POSSIBLE POLLUTANTS · INCOMPLETE COMBUSTION · DIFFERENT PRODUCT BURDEN · CONTROLLED OXYGEN/STEAM CONVERSION · CARBON MONOXIDE + HYDROGEN DOMINANT ROUTE · OUTCOME FIREWALL · PRODUCT IS NOT EFFICIENCY OR CLIMATE PROOF

COMBUSTION → oxidation + useful heat + possible pollutants

INCOMPLETE COMBUSTION → different product burden → GASIFICATION → controlled oxygen/steam conversion → SYNGAS → carbon monoxide + hydrogen dominant route

**CONCEPT / ARCHITECTURE**
- COMBUSTION → oxidation + useful heat + possible pollutants
- INCOMPLETE COMBUSTION → different product burden

**MECHANISM / APPLICATION**
- GASIFICATION → controlled oxygen/steam conversion
- SYNGAS → carbon monoxide + hydrogen dominant route

**READINESS / STATUS LIMIT**
- OUTCOME FIREWALL → product is not efficiency or climate proof
- MUST REMEMBER: Chemistry fundamentals link atomic structure, mole/stoichiometry, bonding,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME FIREWALL → product is not efficiency or climate proof.

10

STAGE 10

## ENVIRONMENTAL COMPARTMENT MAP

ENVIRONMENTAL COMPARTMENT MAP · OZONE PROTECTIVE ROLE · OZONE POLLUTANT ROLE · NUTRIENTS, ACIDS, SALTS, PERSISTENT CHEMICALS · CONCENTRATION + ROUTE + DURATION + CONDITIONS · NOT SAFETY, HAZARD OR REGULATION EVIDENCE · CLOSE DISTINCTION · ATOM IS NOT MOLECULE, MOLE IS NOT MASS, IONIC

STRATOSPHERE → ozone protective role

TROPOSPHERE → ozone pollutant role → WATER / SOIL → nutrients, acids, salts, persistent chemicals → EXPOSURE → concentration + route + duration + conditions

**CONCEPT / ARCHITECTURE**
- STRATOSPHERE → ozone protective role
- TROPOSPHERE → ozone pollutant role

**MECHANISM / APPLICATION**
- WATER / SOIL → nutrients, acids, salts, persistent chemicals
- EXPOSURE → concentration + route + duration + conditions

**READINESS / STATUS LIMIT**
- LABEL → not safety, hazard or regulation evidence
- CLOSE DISTINCTION: Atom is not molecule, mole is not mass, ionic is not covalent,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Atom is not molecule, mole is not mass, ionic is not covalent..

11

STAGE 11

## PYQ AND POLICY EVIDENCE LADDER

PYQ AND POLICY EVIDENCE LADDER · WATER POLARITY · MANY IONIC/POLAR SOLUTES, NOT EVERY SUBSTANCE · PRESENCE DIFFERS FROM EXPOSURE AND REGULATION · MATERIAL CLASS DIFFERS FROM UNIVERSAL PERFORMANCE · NOT DEPLOYMENT, OUTCOME, HAZARD OR LAW · EVIDENCE LIMIT · BALANCE SPECIES, CHARGE AND UNITS; STATE

WATER POLARITY → many ionic/polar solutes, not every substance. → BPA / TRICLOSAN → presence differs from exposure and regulation → HYDROGEL → material class differs from universal performance → PRINCIPLE → PROCESS → APPLICATION → INSTITUTION → PAGE / DOCUMENT → not deployment, outcome, hazard or law

**STARTING CONDITION / INPUT**
- WATER POLARITY → many ionic/polar solutes, not every substance
- BPA / TRICLOSAN → presence differs from exposure and regulation

**SCIENTIFIC OR ENGINEERING MECHANISM**
- HYDROGEL → material class differs from universal performance
- PRINCIPLE → PROCESS → APPLICATION → INSTITUTION

**OUTPUT / FAILURE BOUNDARY**
- PAGE / DOCUMENT → not deployment, outcome, hazard or law
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state.

E

EXTRA

## SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · CHEMISTRY, CATALYSIS, MATERIALS AND MOLECULAR RESEARCH. · ELECTROCHEMISTRY, CORROSION, BATTERIES AND RELATED TRANSLATIONAL TECHNOLOGIES. · METALLURGICAL ECOSYSTEM · EXTRACTION, ALLOY BEHAVIOR AND INDUSTRIAL MATERIALS. · NATIONAL GREEN HYDROGEN MISSION · LINKS ELECTROCHEMISTRY, CATALYSTS, STORAGE AND STANDARDS TO DECARBONIZATION STRATEGY. · UNEP PLASTICS-TREATY PROCESS.

**OPTIONAL SCIENTIFIC / ENGINEERING DEPTH**
- CSIR-NCL: chemistry, catalysis, materials and molecular research.
- CSIR-CECRI: electrochemistry, corrosion, batteries and related translational technologies.
- CSIR-NML / metallurgical ecosystem: extraction, alloy behavior and industrial materials.
- National Green Hydrogen Mission: links electrochemistry, catalysts, storage and standards to decarbonization strategy.

**READINESS, UNIT OR STATUS LIMIT**
- UNEP plastics-treaty process: international governance response to polymer lifecycle issues.
- Fertilizer chemistry is central to food security, soil management and subsidy debates.
- Electrochemistry matters for batteries, plating, corrosion control and hydrogen pathways.
- Polymer chemistry shapes packaging, healthcare disposables, pipes, textiles and waste burdens.

**COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE**
- Metallurgy underlies infrastructure, railways, defence, power systems and manufacturing competitiveness.
- Biomolecule chemistry supports diagnostics, therapeutics and industrial biotechnology.
- CAUTION: Limitation 1: chemical utility and environmental externality often move together
- — the same polymer that lowers cost may raise waste-management burden.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels